Chapter Eight

Scrum

Introduction to Scrum

- Scrum is one of the most popular agile methodology
- It is a lightweight, iterative and incremental framework.
- It breaks down the development phases into stages or cycles called "**sprints**".
- Scrum allows us to rapidly and repeatedly inspect actual working software (every two weeks to one month).

Scrum…

- Scrum is an agile process that allows the developers to focus on delivering the highest business value in the shortest time.
- In scrum the group of people exert their forces and initiatives towards achieving the common goal.
- The team members manage each other to determine the best way to deliver the highest priority features.
- Every two weeks to a month anyone can see real working software and decide to release it as is or continue to enhance for another iteration.

Historical Background

- Jeff Sutherland
  - first person to apply concepts of Scrum to software development in 1993
- A variation of Sashimi
  - Japanese designed an all at once approach after their bad experiences with Waterfall
- The initial use of the word "Scrum" was used in 1987 to describe time-boxed, self-organizing teams in product development

Historical Background (cont.)

- Jeff Sutherland and Ken Schwaber
  - collaborated to define the process through 1995. In 1996 wrote the seminal article for Scrum Software Development process
  - jointly used and improved Scrum at a variety of software development organizations from 1996 until now.

Description Overview

- Scrum is named after the game of Rugby in which a group is responsible for picking up the ball and moving it forward.
- It is an iterative, incremental process for developing any product or managing any work.
- Scrum focuses on the entire organization for its implementation to be a success.

Characteristics

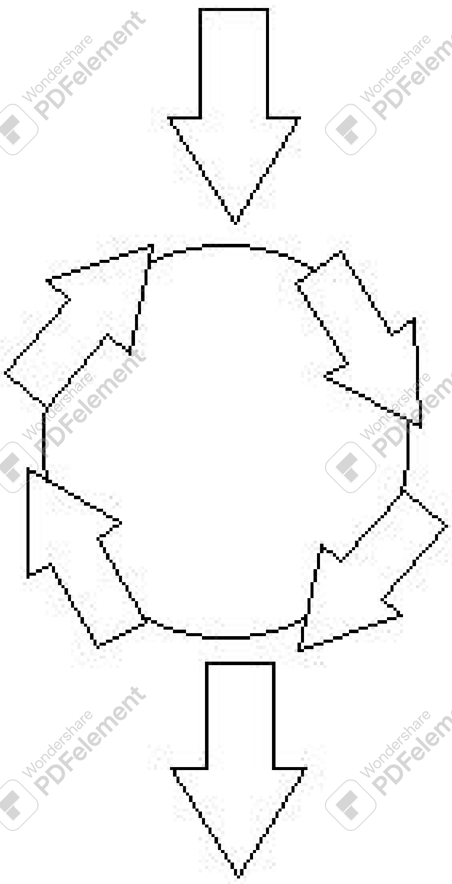
- Self-organizing and cross-functional teams
- Product progresses in a series of month-long “sprints”
- Requirements are captured as items in a list of “product backlog”
- Uses generative rules to create an agile environment for delivering projects

Scrum principles

- Scrum principles include:
  - **Quality work**: empowers everyone involved to be feel good about their job.
  - **Assume Simplicity**: Scrum is a way to detect and cause removal of anything that gets in the way of development.
  - **Embracing Change**: Team based approach to development where requirements are rapidly changing.
  - **Incremental changes**: Scrum makes this possible using sprints where a team is able to deliver a product (iteration) deliverable within 30 days.

Scrum Methodology

- **Pregame**
  - Planning
  - System Architecture/High Level Design
- **Game**
  - Sprints (Concurrent Engineering)
  - Develop (Analysis,Design,Develop)
  - Wrap
  - Review
  - Adjust
- **Postgame**
  - Closure



Description Components

Processes:

- **<u>Pregame</u>**

  Planning and Architecture:

  - Identify project
  - Prioritizing functional requirements
  - Identify resources available
  - Establishing the target environment

Description Components (cont.)

Processes:

- **<u>Game</u>**

  Sprints:- lasts for 30 days

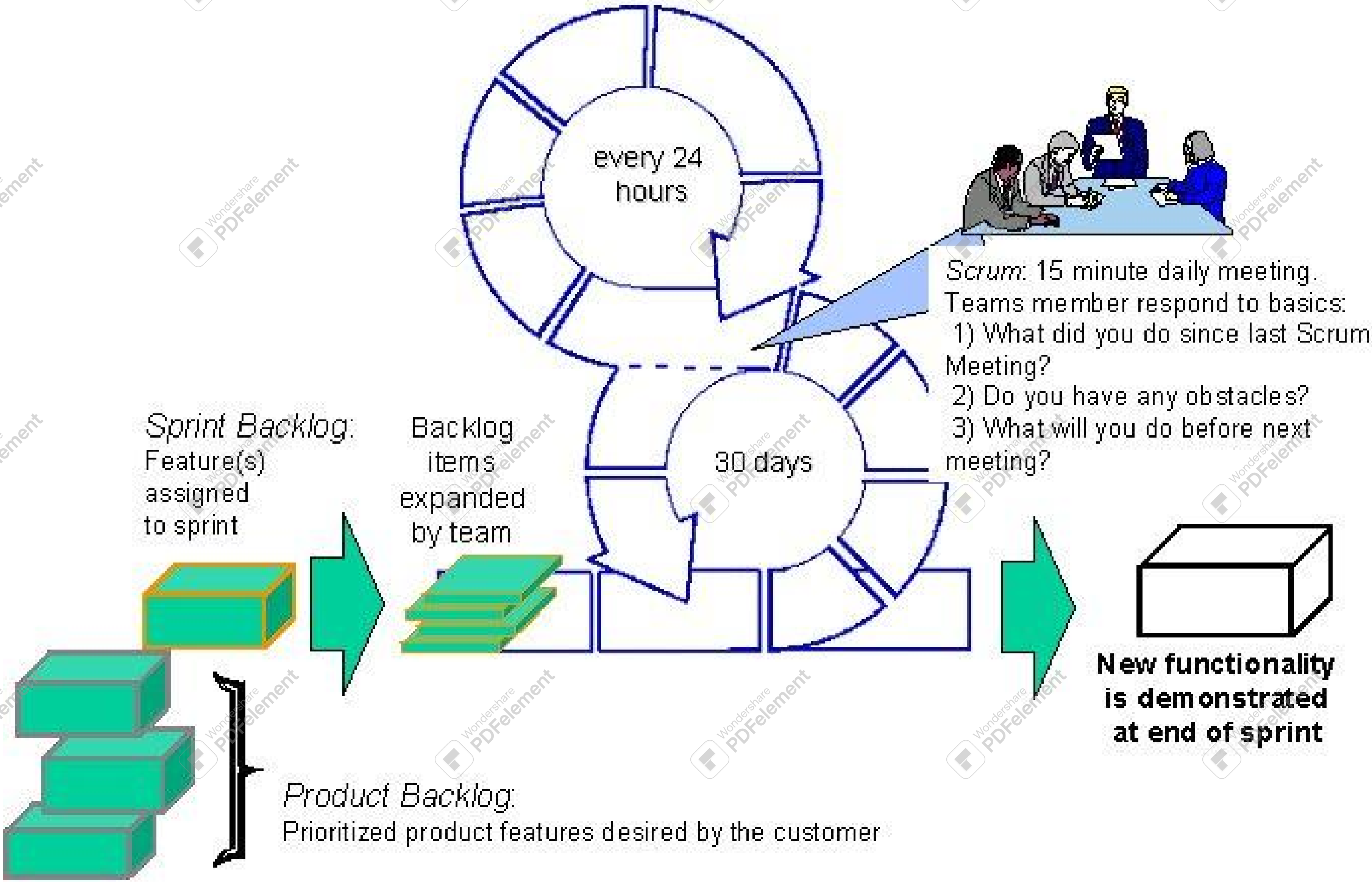
  - Analysis, Design, Develop
  - Testing (this happens throughout sprint)
  - Review
  - Adjust

- **<u>Postgame</u>**

  - Closure (this includes delivering a functioning deliverable, sign-off, start next sprint).

Description Components (cont.)

- Values:
  - Flexible deliverable
  - Flexible schedule
  - Small teams
  - Frequent reviews
  - Collaboration
  - Team Empowerment
  - Adaptability


every 24 hours
Scrum: 15 minute daily meeting.
Teams member respond to basics:
1) What did you do since last Scrum Meeting?
2) Do you have any obstacles?
3) What will you do before next meeting?
Sprint Backlog:
Feature(s) assigned to sprint
Backlog items expanded by team
30 days
New functionality is demonstrated at end of sprint
Product Backlog:
Prioritized product features desired by the customer

Description Advantages

- Extreme value - reduces risks
- Supports business value driven software development.
- Control of very complex process of product development
- Allows Developers to focus on delivering a usable functionality to the client
- Generates productivity improvements by implementing a framework that empowers teams and thrives on change

Description Advantages (cont.)

- Insists that the Client prioritize required functionality.
- Ability to respond to the unpredictable in any project requirements.
- Flexibility
- Knowledge sharing between Developers
- Collective ownership
- Supports an “Object Oriented” technology

Description Disadvantages

- Scrum is not effective for small projects
- Expensive to implement
- Training is required

Usage Guidelines – When to use

- Organizations are willing to do anything and everything for a project to succeed
- Work is delivered in increments
- Work is measured and controlled
- Requirements are not clearly defined.
- Productivity is maximized by applying known technologies

Usage Guidelines – When to avoid

- There isn't a flexible environment
- Corporate culture isn't conducive to this of development environment
- Teams of developers are more than 10. Six is ideal.
- Cost is a major issue
- No management support
- No formal training available

Usage Guidelines – Implementation

- Need for an extra member just in case an active member is absent.
- Location: Although not impossible, its hard to implement Scrum when all team members are not in the same location.

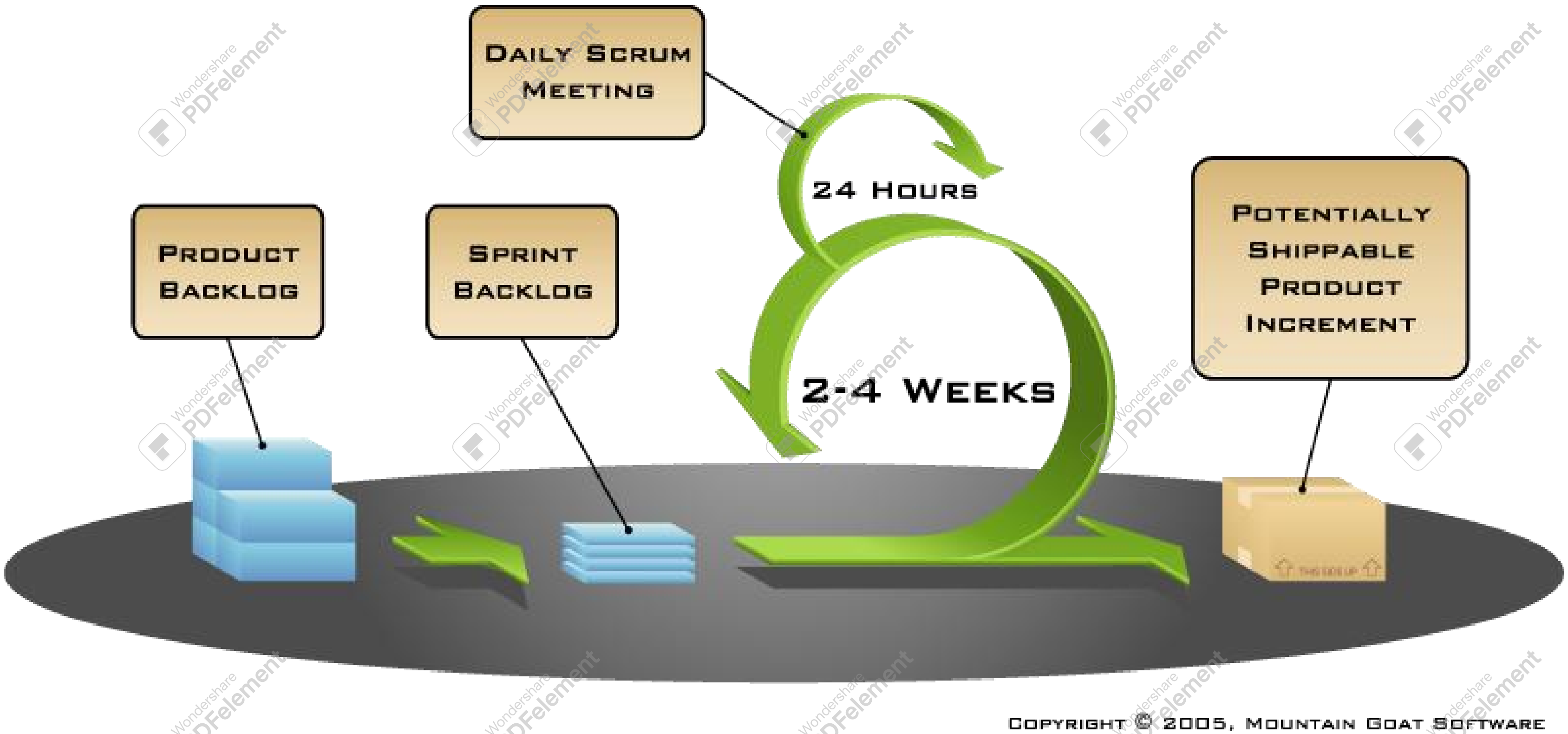
CMM vs. Agile

- Capability Maturity Model(CMM) is a process improvement approach aimed on the organizational improvement.
- It observes existing behaviors and does no planning while agile is creating new processes and functions around detailed plans to deliver shippable products.

Scrum and CMM

- CMM advocates Repeated Defined problems, solutions, Developers and organizational environment.
- Scrum says that this is not entirely possible because developers change from one project to another.
- Scrum assumes that the development process is always empirical and not defined.
- Scrum says uncertainties are impossible to measure, therefore, looks beyond the repeatable /defined approach.

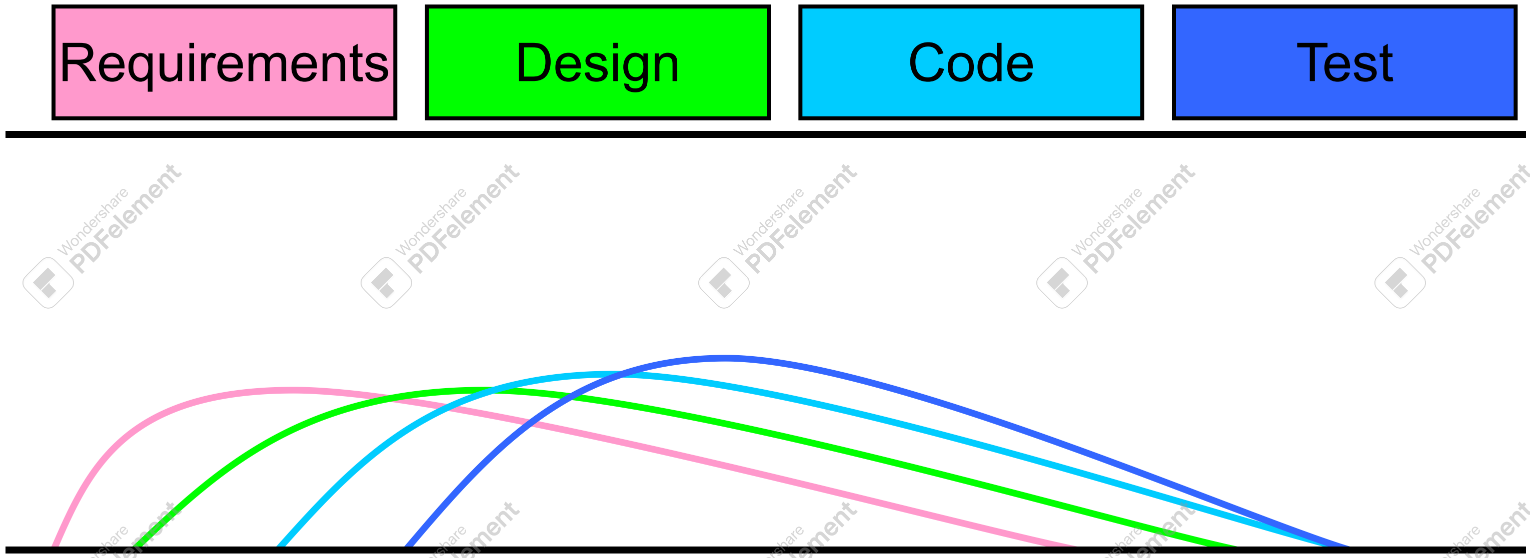
How Scrum Works?



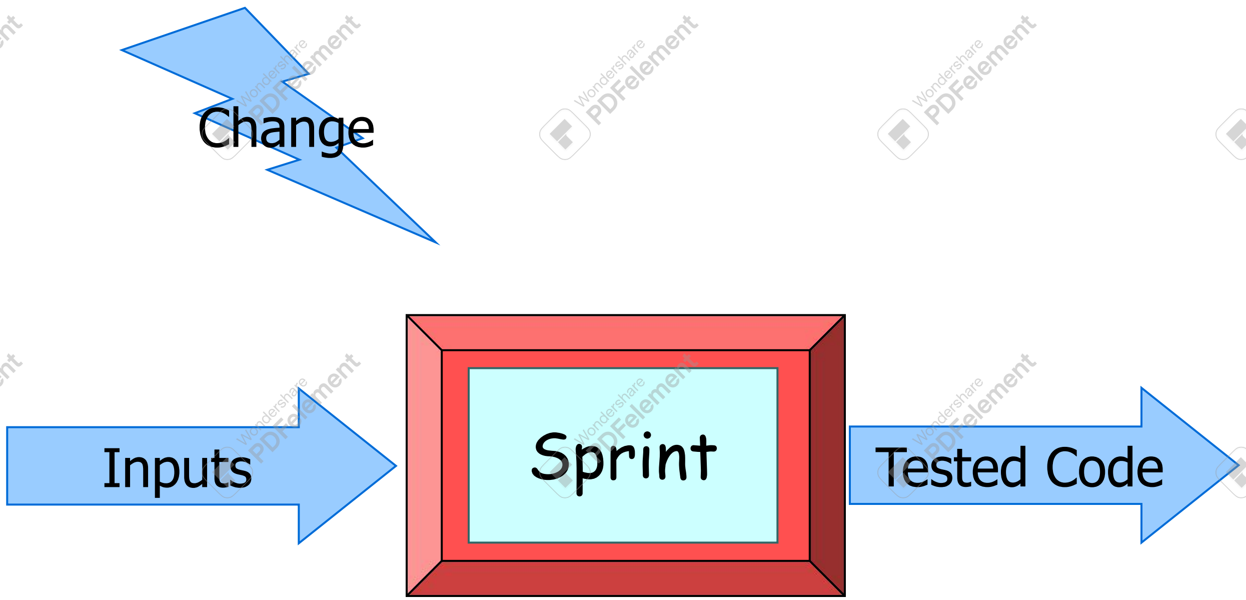
Sprints

- Scrum projects make progress in a series of "sprints"
  - Analogous to XP iterations
- Target duration is one month
- Product is designed, coded, and tested during the sprint

Sequential vs. Overlapping Development



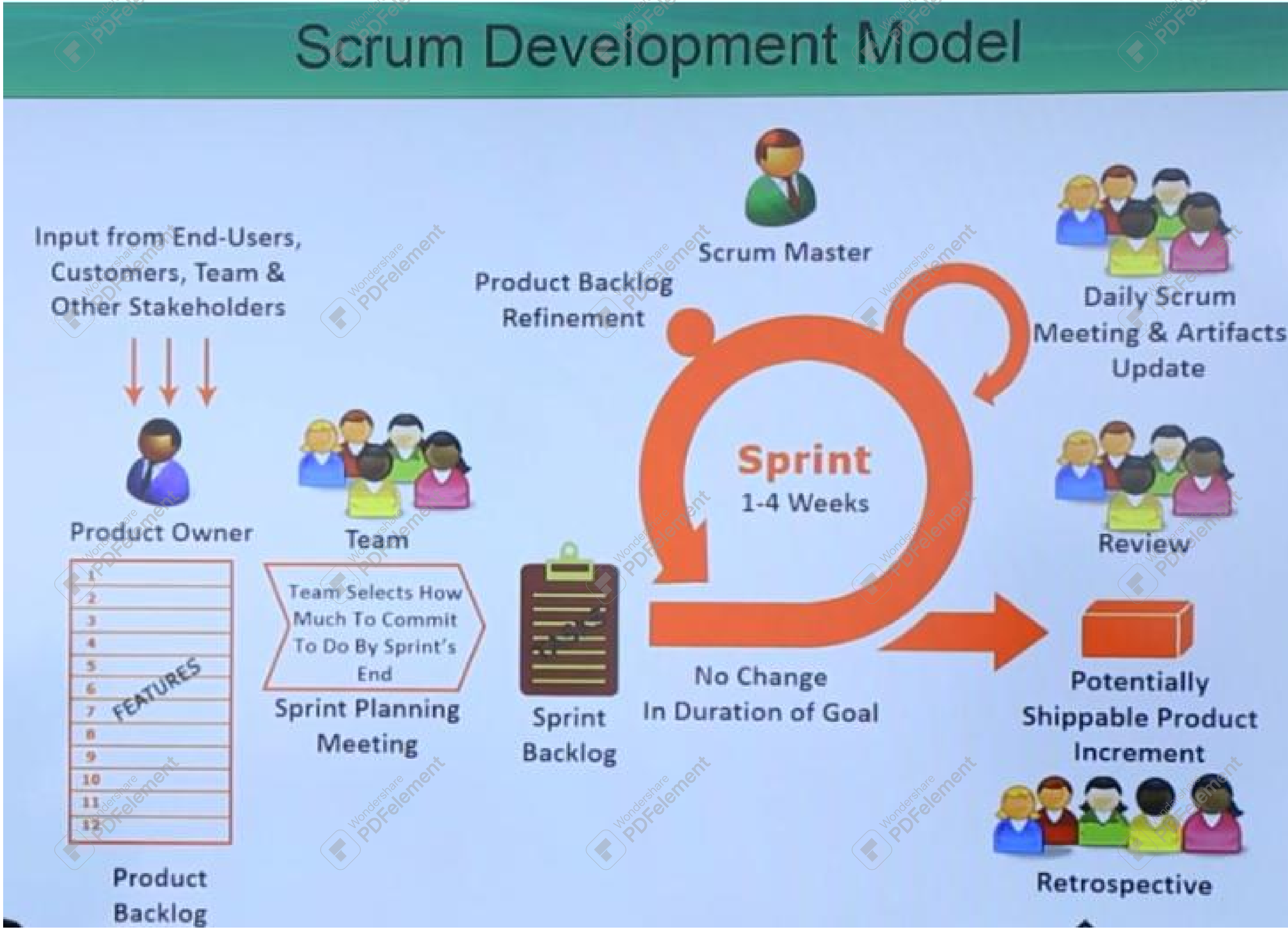
No changes during the sprint



- Plan sprint durations around how long you can commit to keeping change out of the sprint

Scrum Framework

- **Roles** : Product Owner, ScrumMaster, Team
- **Ceremonies** : Sprint Planning, Sprint Review, Sprint Retrospective, & Daily Scrum Meeting
- **Artifacts** : Product Backlog, Sprint Backlog, and Burndown Chart


Scrum Development Model
Input from End-Users, Customers, Team & Other Stakeholders
Product Owner
Team
Scrum Master
Product Backlog Refinement
Daily Scrum Meeting & Artifacts Update
Sprint
1-4 Weeks
Review
FEATURES
1
2
3
4
5
6
7
8
9
10
11
12
Team Selects How Much To Commit To Do By Sprint's End
Sprint Planning Meeting
Sprint Backlog
No Change In Duration of Goal
Potentially Shippable Product Increment
Retrospective
Product Backlog

Product Owner

- Define the features of the product
- Decide on release date and content
- Be responsible for the profitability of the product
- Prioritize features according to market value
- Adjust features and priority every iteration, as needed
- Accept or reject work results.

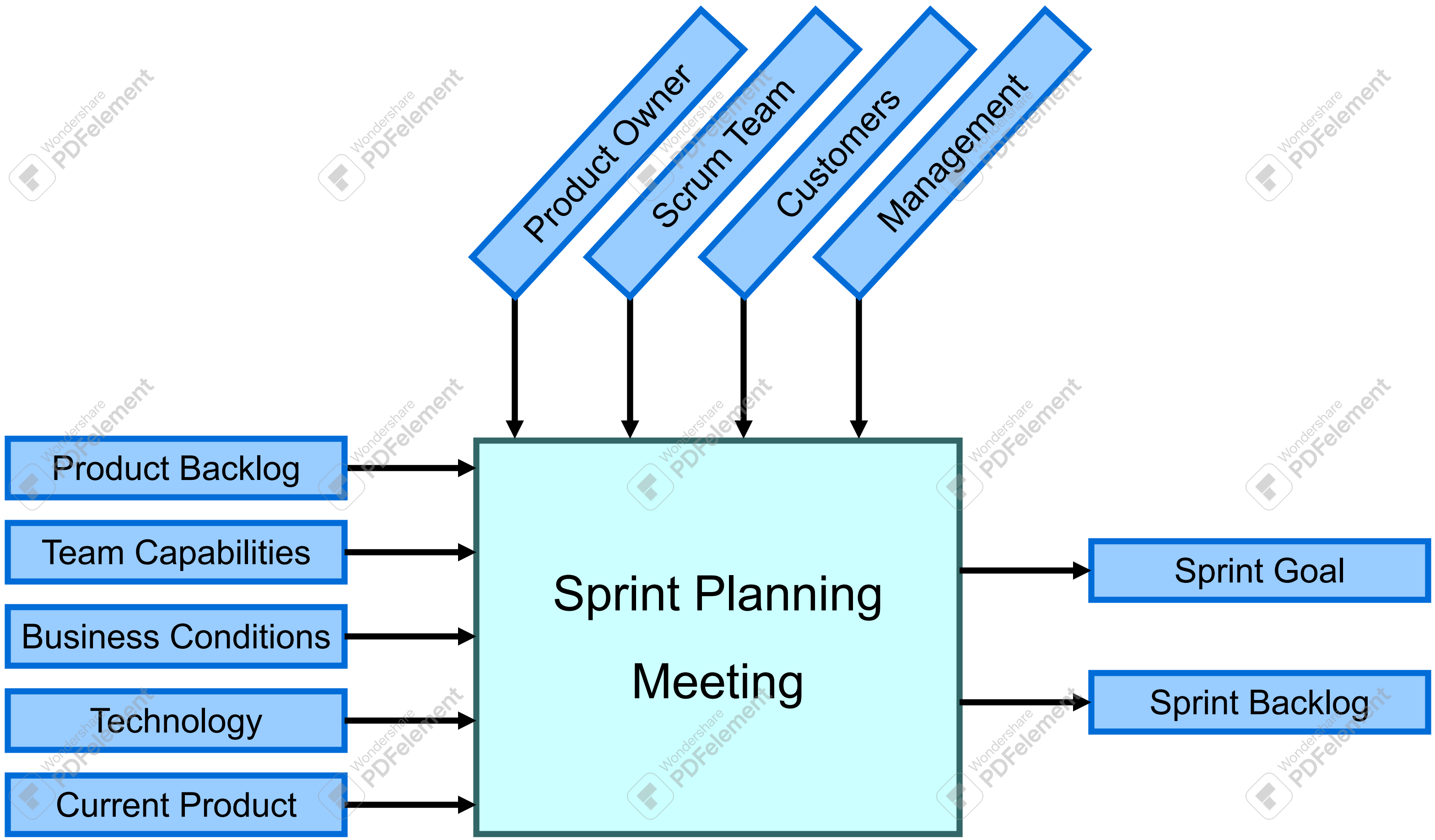
The Scrum Master

- Represents management to the project
- Responsible for enacting Scrum values and practices
- Removes obstacles
- Ensure that the team is fully functional and productive
- Enable close cooperation across all roles and functions
- Shield the team from external interferences

Scrum Team

- Typically 5-10 people
- Cross-functional
  - QA, Programmers, UI Designers, etc.
- Members should be full-time
  - May be exceptions (e.g., System Admin, etc.)
- Teams are cross-functional
- Membership can change only between sprints

Sprint Planning Meeting



Parts of Sprint Planning Meeting

- 1st Part:
  - Creating Product Backlog
  - Determining the Sprint Goal.
  - Participants: Product Owner, Scrum Master, Scrum Team
- 2nd Part:
  - Participants: Scrum Master, Scrum Team
  - Creating Sprint Backlog

Pre-Project/Kickoff Meeting

- A special form of Sprint Planning Meeting
- Meeting before the begin of the Project

Sprint

- A month-long iteration, during which is incremented a product functionality
- No outside influence can interfere with the Scrum team during the Sprint
- Each Sprint begins with the Daily Scrum Meeting

Daily Scrum

- Parameters
  - Daily
  - 15-minutes
  - Not for problem solving
  - Help avoid other unnecessary meetings
- Three questions:
  1. What did you do yesterday
  2. What will you do today?
  3. What obstacles are in your way?

Daily Scrum

- Is NOT a problem solving session
- Is NOT a way to collect information about WHO is behind the schedule
- Is a meeting in which team members make commitments to each other and to the Scrum Master
- Is a good way for a Scrum Master to track the progress of the Team

Sprint Review Meeting

- Team presents what it accomplished during the sprint
- Typically takes the form of a demo of new features or underlying architecture
- Participants
  - Customers
  - Management
  - Product Owner
  - Other engineers

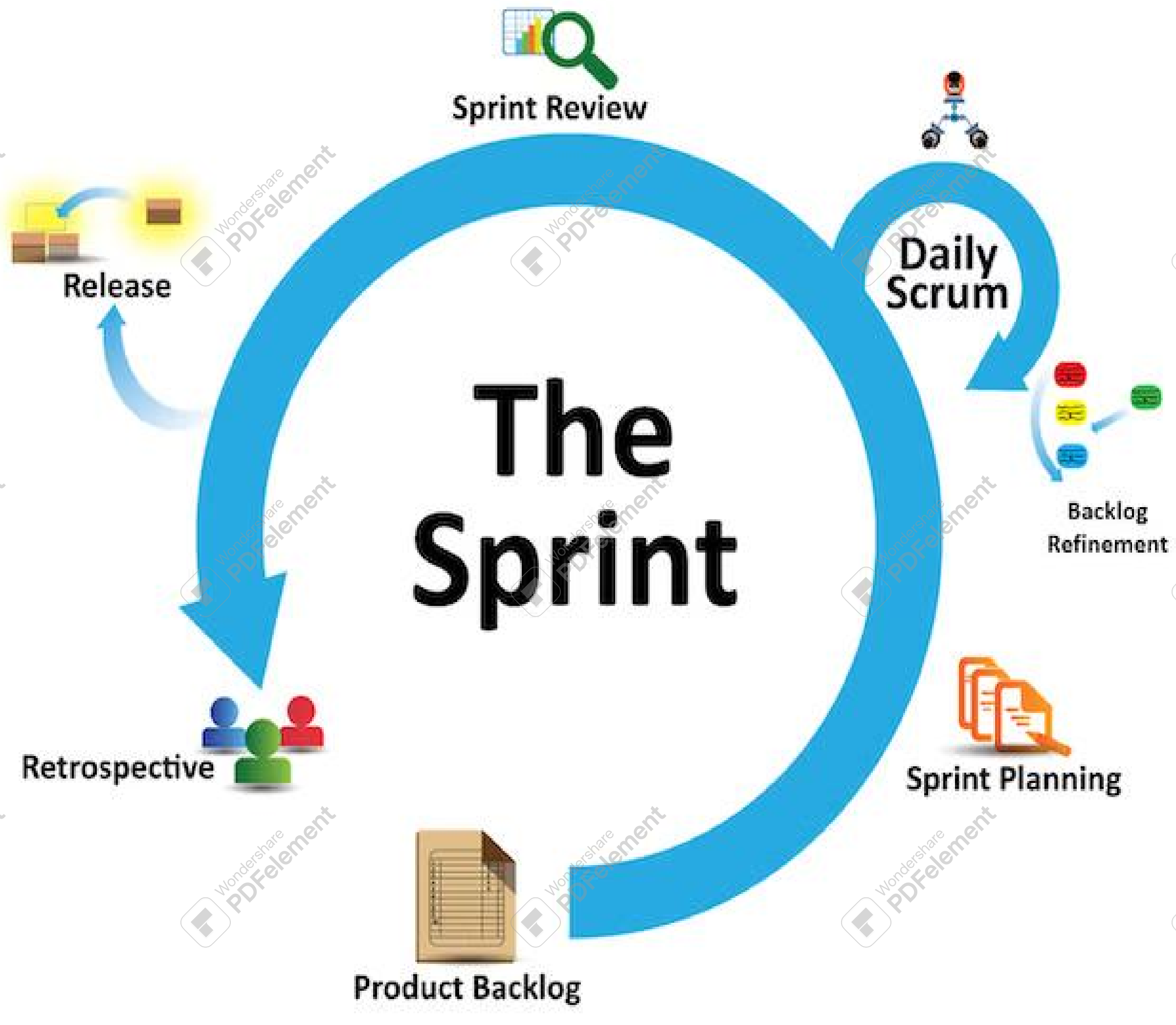
Sprint Retrospective Meeting

- The Sprint Retrospective is an opportunity for the Scrum team to look at itself and create a plan for improvements to be enacted during the next sprint
- Scrum Team only
- Feedback meeting
- Three questions
  - Start
  - Stop
  - Continue

Sprint Reviews vs. Sprint Retrospectives

- Sprint Review focuses on improving so the team can deliver a better product, whereas a Sprint Retrospective focuses on improving the overall system so the team can work more harmoniously and find flow together.
- To create an effective, functional, technically sound, and user-centric product, you need all stakeholders and developers to align in the sprint review meeting.

Cont…



Sprint Review
Daily
Scrum
Release
The
Sprint
Backlog
Refinement
Retrospective
Sprint Planning
Product Backlog

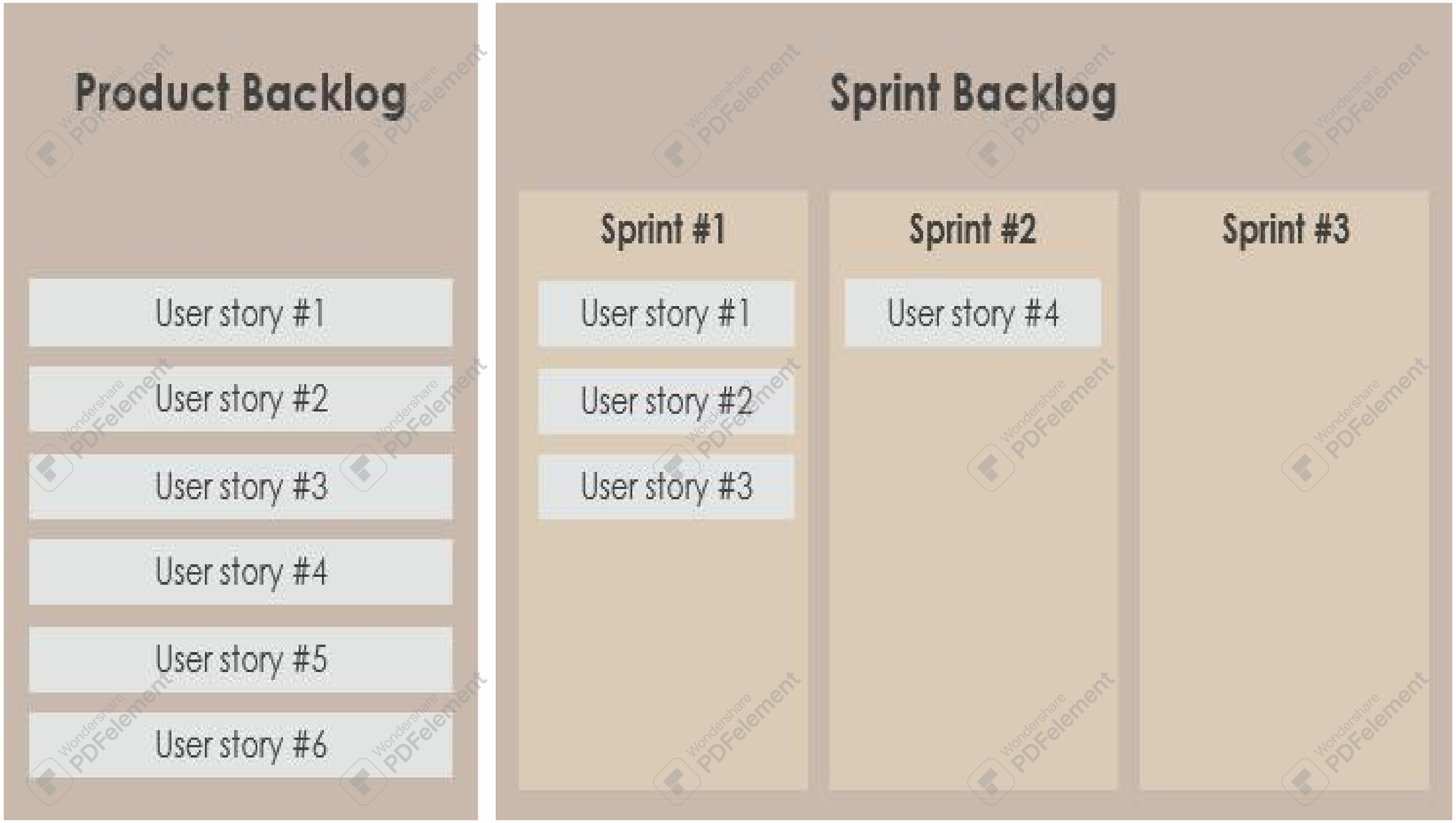
Product Backlog

- A list of all desired work on the project
- It is a prioritized list of work for the development team that is derived from the roadmap and its requirements .
- List is prioritized by the Product Owner
  - Typically a Product Manager, Marketing, Internal Customer, etc.

Product Backlog

- Requirements for a system, expressed as a prioritized list of Backlog Items
- Is managed and owned by a Product Owner
- Spreadsheet (typically)
- Usually is created during the Sprint Planning Meeting
- Can be changed and re-prioritized whenever necessary

Product vs. Sprint backlog example



Sample Product Backlog

	Item #	Description	Est	By
Very High				
	1	**Finish database versioning**	16	KH
	2	**Get rid of unneeded shared Java in database**	8	KH
	-	**Add licensing**	-	-
	3	Concurrent user licensing	16	TG
	4	Demo / Eval licensing	16	TG
		Analysis Manager		
	5	File formats we support are out of date	160	TG
	6	Round-trip Analyses	250	MC
High				
	-	**Enforce unique names**	-	-
	7	In main application	24	KH
	8	In import	24	AM
	-	**Admin Program**	-	-
	9	Delete users	4	JM
	-	**Analysis Manager**	-	-
	10	When items are removed from an analysis, they should show up again in the pick list in lower 1/2 of the analysis tab	8	TG
	-	**Query**	-	-
	11	Support for wildcards when searching	16	T&A
	12	Sorting of number attributes to handle negative numbers	16	T&A
	13	Horizontal scrolling	12	T&A
	-	**Population Genetics**	-	-
	14	Frequency Manager	400	T&M
	15	Query Tool	400	T&M
	16	Additional Editors (which ones)	240	T&M
	17	Study Variable Manager	240	T&M
	18	Haplotypes	320	T&M
	19	**Add icons for v1.1 or 2.0**	-	-
	-	**Pedigree Manager**	-	-
	20	Validate Derived kindred	4	KH
Medium				
	-	**Explorer**	-	-
	21	Launch tab synchronization (only show queries/analyses for logged in users)	8	T&A
	22	Delete settings (?)	4	T&A

From Sprint Goal to Sprint Backlog

- Scrum team takes the Sprint Goal and decides what tasks are necessary
- Team self-organizes around how they'll meet the Sprint Goal
  - Manager doesn't assign tasks to individuals
- Managers don't make decisions for the team

Sprint Backlog during the Sprint

- Changes
  - Team adds new tasks whenever they need to in order to meet the Sprint Goal
  - Team can remove unnecessary tasks
  - But: Sprint Backlog can only be updated by the team
- Estimates are updated whenever there's new information

Sprint Backlog

- A subset of Product Backlog Items, which define the work for a Sprint
- Is created ONLY by Team members
- Each Item has it's own status
- Should be updated every day

Sprint Backlog

- No more than 300 tasks in the list
- If a task requires more than 16 hours, it should be broken down
- Team can add or subtract items from the list.
- Product Owner is not allowed to do it

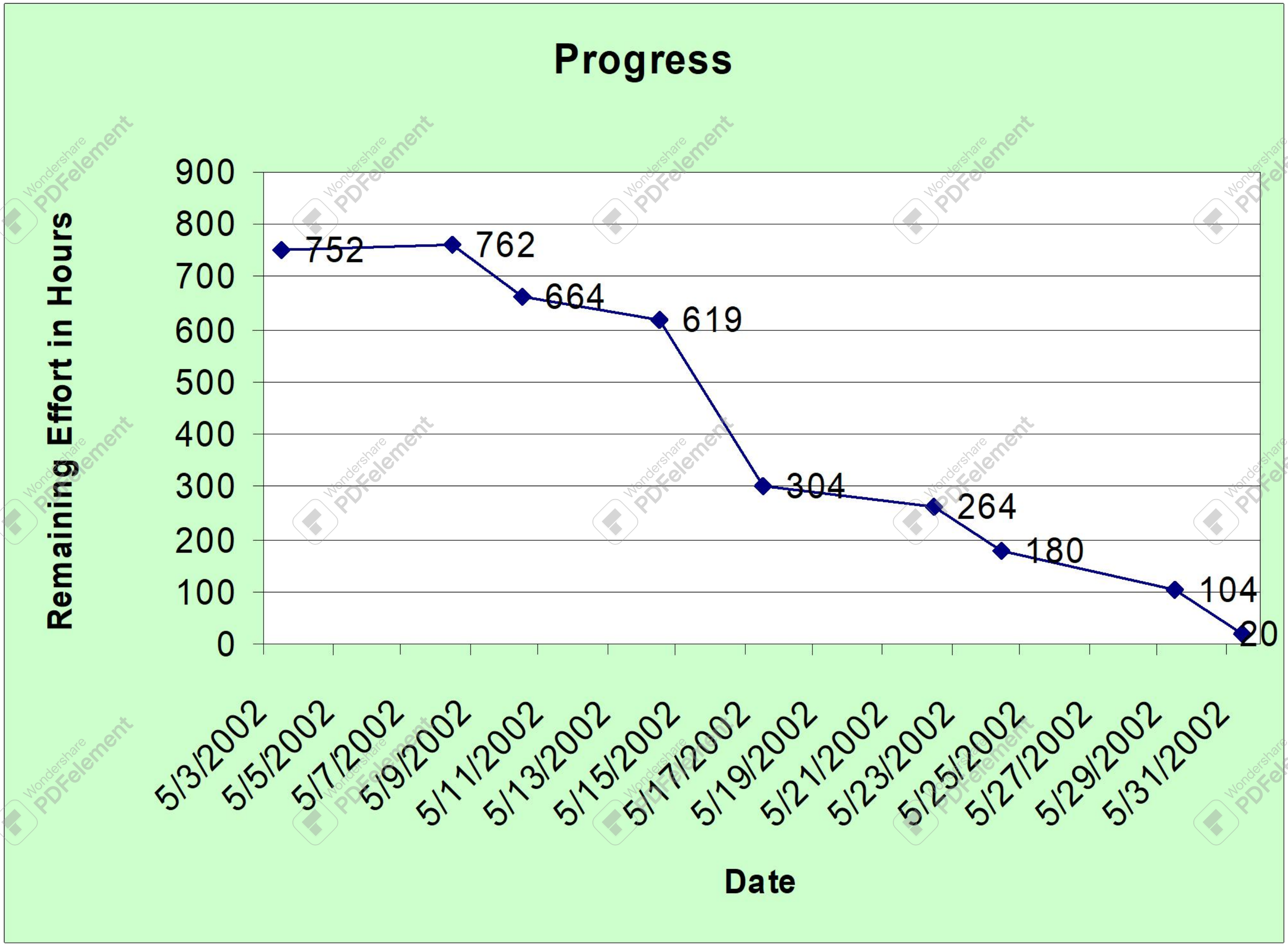
Sample Sprint Backlog

Who	Description	7/22/2002	7/24/2002	7/26/2002	7/31/2002	
	Days Left in Sprint	**15**	**13**	**10**	**8**	
	Total Estimated Hours:	**554**	**458**	**362**	**270**	**0**
-	**User's Guide**	-	-	-	-	-
SM	Start on Study Variable chapter first draft	16	16	16	16	
SM	Import chapter first draft	40	24	6	6	
SM	Export chapter first draft	24	24	24	6	
	Misc. Small Bugs					
JM	Fix connection leak	40				
JM	Delete queries	8	8			
JM	Delete analysis	8	8			
TG	Fix tear-off messaging bug	8	8			
JM	View pedigree for kindred column in a result set	2	2	2	2	
AM	Derived kindred validation	8				
	Environment					
TG	Install CVS	16	16			
TBD	Move code into CVS	40	40	40	40	
TBD	Move to JDK 1.4	8	8	8	8	
	Database					
KH	Killing Oracle sessions	8	8	8	8	
KH	Finish 2.206 database patch	8	2			
KH	Make a 2.207 database patch	8	8	8	8	
KH	Figure out why 461 indexes are created	4				

Sprint Burn down Chart

- Depicts the total Sprint Backlog hours remaining per day
- Shows the estimated amount of time to release
- Ideally should burn down to zero to the end of the Sprint
- Actually is not a straight line

Sprint Burndown Chart



Release Burndown Chart

- Will the release be done on right time?
- X-axis: sprints
- Y-axis: amount of hours remaining
- The estimated work remaining can also burn up

Scalability of Scrum

- A typical Scrum team is 6-10 people
- Jeff Sutherland - up to over 800 people
- "Scrum of Scrums" or what called "Meta-Scrum“
- Frequency of meetings is based on the degree of coupling between packets

Pros/Cons

- *Advantages*
  - *Completely developed and tested features in short iterations*
  - *Simplicity of the process*
  - *Clearly defined rules*
  - *Increasing productivity*
  - *Self-organizing*
  - *each team member carries a lot of responsibility*
  - *Improved communication*
  - *Combination with Extreme Programming*
- *Drawbacks*
  - *“Undisciplined hacking” (no written documentation)*
  - *Violation of responsibility*
  - *Current mainly carried by the inventors*

Thank You!!